

BASIC ELECTRICAL ENGINEERING LAB								
I B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE02	ESC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	30	70	75
COURSE OBJECTIVES:								
The course should enable the students to:								
1. Get an exposure to common electrical components and their ratings. 2. Make electrical connections by wires of appropriate ratings. 3. Understand the usage of common electrical measuring instruments. 4. Understand the basic characteristics of transformers and electrical machines.								
COURSE OUTCOMES:								
1. Get an exposure to basic electrical laws. 2. Demonstrate different types of electrical circuits to different excitations. 3. Calculate the relation between the basic electrical parameters. 4. Analyze the basic characteristics of transformers and electrical machines.								
LIST OF EXPERIMENTS								
WEEK-1	INTRODUCTION AND USE OF MEASURING INSTRUMENTS & SAFETY PRECAUTIONS							
To study the usage of electrical instruments and the required precautions to be taken.								
WEEK-2	KIRCHOFF'S LAWS(KVL & KCL)							
To verify kvl and kcl								
WEEK-3	SUPERPOSITION THEOREM							
To verify superposition theorem								
WEEK-4	THEVENIN'S AND NORTON'S THEOREM							
To obtain equivalent circuit of a complex network								
WEEK-5	STEADY STATE AND TRANSIENT RESPONSE OF R-L, R-C & R-L-C CIRCUITS							
To find the steady and transient response of r-l, r-c & r-l-c circuits for step input.								
WEEK-6	OPEN CIRCUIT, SHORT CIRCUIT & LOAD TEST ON SINGLE PHASE TRANSFORMER							
To calculate the efficiency of single phase transformer.								
WEEK-7	CUT OUT VIEW OF DC MACHINE							
Demonstration on constructional and cut out view of dc machine								
WEEK-8	CUT OUT VIEW OF INDUCTION MOTOR							
Demonstration on constructional and cut out view of single phase induction motor								
WEEK-9	MAGNETIZATION CHARACTERISTICS OF DC SHUNG GENERATOR							
To find the magnetization characteristics of dc shunt generator								
WEEK-10	BRAKE TEST ON DC SHUNT MOTOR							
To find the torque-speed characteristics of dc shunt motor.								
WEEK-11	THREE PHASE TRANSFORMER CONNECTIONS							
To calculte the relation between phase and line voltages								
WEEK-12	BRAKE TEST ON 3-PHASE INDUCTION MOTOR							
To find the torque-slip characteristics of induction motor								
REFERENCE BOOKS:								

1. Department Lab Manual
2. Chakrabarthi, " Circuit Theory", Dhanpat Rai Publications, 6th Edition,2006
3. V K Mehta, Rohit Mehta, "Principles of Electrical Machines", S Chand Publications, 1st Edition,2006
4. Nagrath & DP Kothari, "Electrical Machines", Mcgraw Hill Education Publications, 4th Edition, 2010.

WEB REFERENCES:

1. <http://www.ee.iitkgp.ac.in>
2. <http://www.citchennai.edu.in>